

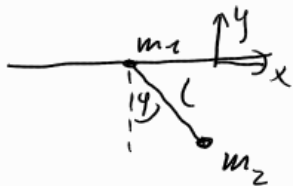
$$L = T - V \quad \text{Lagrangian}$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = 0$$

$$H = \sum_i \frac{\partial L}{\partial \dot{q}_i} \dot{q}_i - L = T + V \quad \text{Hamiltonian}$$

$$\frac{\partial L}{\partial t} = 0 \Rightarrow \frac{dH}{dt} = 0$$

① mat. nihalo na premičnem pritrdišču



$$\text{vezi: } y_1 = 0$$

$$l = \sqrt{(x_2 - x_1)^2 + y_2^2} = \text{konst.}$$

$$x_1, y_1, x_2, y_2$$

→ rabimo 2 pospli koord. (ker imamo 2 vezi)

$$x_1, \varphi$$

$$x_2 = x_1 + l \sin \varphi$$

$$y_2 = -l \cos \varphi$$

$$T = \frac{1}{2} m_1 \dot{x}_1^2 + \frac{1}{2} m_2 (\dot{x}_2^2 + \dot{y}_2^2) = \frac{1}{2} m_1 \dot{x}_1^2 + \frac{1}{2} m_2 (\dot{x}_1^2 + 2l \dot{x}_1 \cos \varphi \dot{\varphi} + l^2 \cos^2 \varphi \dot{\varphi}^2 + l^2 \sin^2 \varphi \dot{\varphi}^2)$$

$$V = m_2 g y_2 = -m_2 g l \cos \varphi$$

EL:

$$x_1: \frac{d}{dt} [m_1 \dot{x}_1 + m_2 \dot{x}_1 + m_2 l \cos \varphi \dot{\varphi}] - 0 = 0$$

$$\varphi: \frac{d}{dt} [m_2 l \dot{x}_1 \cos \varphi + m_2 l^2 \dot{\varphi}] - [m_2 \dot{x}_1 (-\sin \varphi) \dot{\varphi} + m_2 g l (-\sin \varphi)] = 0$$

$$l \ddot{\varphi} + \ddot{x} \cos \varphi + \cancel{l \dot{x} \sin \varphi \dot{\varphi}} + \cancel{l \dot{x}_1 \sin \varphi \dot{\varphi}} + g \sin \varphi = 0$$

$$l \ddot{\varphi} + \ddot{x} \cos \varphi + g \sin \varphi = 0$$

$$m_1 \dot{x}_1 + m_2 \dot{x}_1 + m_2 l \cos \varphi \dot{\varphi} = p_x = \text{konst.}$$

$$\dot{p}_x = (m_1 + m_2) \ddot{x}_1 + m_2 l (\cos \varphi \ddot{\varphi} - \sin \varphi \dot{\varphi}^2) = 0$$

$$\ddot{x}_1 = \frac{m_2}{m_1 + m_2} l (\sin \varphi \dot{\varphi}^2 - \cos \varphi \ddot{\varphi})$$

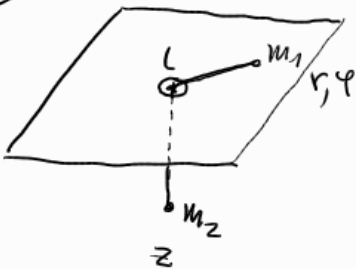
$$l \ddot{\varphi} + \frac{m_2}{m_1+m_2} l (\sin\varphi \cos\varphi \dot{\varphi}^2 - \cos^2\varphi \ddot{\varphi}) + g \sin\varphi = 0$$

$$\ddot{\varphi} l \left(1 - \frac{m_2}{m_1+m_2} \cos^2\varphi\right) + \frac{m_2}{m_1+m_2} l \sin\varphi \cos\varphi \dot{\varphi}^2 + g \sin\varphi = 0 \quad (\varphi \ll 1)$$

$$\ddot{\varphi} l \left(1 - \frac{m_2}{m_1+m_2}\right) + g \varphi = 0$$

$$\ddot{\varphi} + \frac{g}{l} \frac{m_1+m_2}{m_1} \varphi = 0 \rightarrow \omega = \sqrt{\frac{g}{l} \frac{m_1+m_2}{m_1}} \quad \text{v limiti } m_1 \gg m_2: \omega = \sqrt{\frac{g}{l}}$$

(2)



$$T = \frac{m_1}{2} (\dot{r}^2 + r^2 \dot{\varphi}^2) + \frac{m_2}{2} \dot{z}^2 = \frac{m_1}{2} (\dot{r}^2 + r^2 \dot{\varphi}^2) + \frac{m_2}{2} \dot{r}^2$$

$$V = -m_2 g z = m_2 g (l - r)$$

$$\text{vez: } -z + r = l \rightarrow z = r - l$$

$$r: \frac{d}{dt} [m_1 \dot{r} + m_2 \dot{r}] - [m_1 r \dot{\varphi}^2 + m_2 g] = 0 \rightarrow (m_1 + m_2) \ddot{r} - m_1 r \dot{\varphi}^2 + m_2 g = 0$$

$$\varphi: \frac{d}{dt} [m_1 r^2 \dot{\varphi}] = 0 \rightarrow \dot{\varphi} = \frac{p_\varphi}{m_1 r^2}$$

$$(m_1 + m_2) \ddot{r} - \frac{p_\varphi^2}{m_1 r^3} + m_2 g = 0$$

(1) $r = \text{konst.}$

$$p_\varphi^2 = m_1 m_2 g r_0^3 = \dot{\varphi}^2 m_1^2 r_0^4 \rightarrow \dot{\varphi}^2 = \frac{m_2}{m_1} \frac{g}{r_0}$$

(2) $r = r_0 + \xi$

$$\ddot{\xi} (m_1 + m_2) + \frac{m_2}{2} g - \frac{p_\varphi^2}{m_1 r_0^3} \left(1 - \frac{3\xi}{r_0}\right) = 0$$

$$\ddot{\xi} - \underbrace{\xi \frac{3 p_\varphi^2}{r_0^4 m_1 (m_1 + m_2)}}_{\omega^2} = 0$$

$$\omega^2 = \frac{3 p_\varphi^2 \omega_0^2 m_1^2 r_0^4}{r_0^4 m_1 (m_1 + m_2)} = \frac{m_1}{m_1 + m_2} 3 \omega_0$$

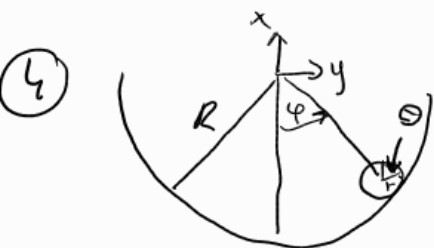


3) Gibanje po polkrogli brez trenja



kdaj se odlepi od podlage?

$$\cos \Theta = \frac{2}{3}$$



$$V = -(R-r) \cos \varphi mg$$

$$T = \frac{1}{2} m ((R-r) \dot{\varphi})^2 + \frac{1}{2} \frac{5}{2} m r^2 \dot{\Theta}^2$$

$$\text{vez: } r d\Theta = R d\varphi$$

$$r \dot{\Theta} = R \dot{\varphi}$$

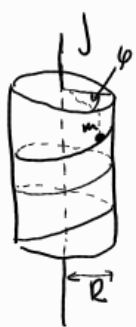
$$T = \frac{1}{2} m (R-r)^2 \dot{\varphi}^2 + \frac{5}{4} m R^2 \dot{\varphi}^2$$

$$\varphi: \frac{d}{dt} \left[\frac{1}{2} m (R-r)^2 \dot{\varphi} + \frac{5}{2} m R^2 \dot{\varphi} \right] + (R-r) (-\sin \varphi) mg = 0$$

$$\ddot{\varphi} \left((R-r)^2 + \frac{5}{2} R^2 \right) - (R-r) g \sin \varphi = 0 \quad / \varphi \ll$$

$$\omega = \sqrt{\frac{(R-r)g}{(R-r)^2 + \frac{5}{2} R^2}}$$

→ mat. nihalo z efektivno dolžino $l_{\text{ef}} = \left(R-r + \frac{5}{2} \right) \frac{R^2}{R-r}$



masa gledana valj

$$z = z_0 + p(\varphi - \phi)$$

↑ valj

$$m: x = R \sin \varphi$$

$$y = R \cos \varphi$$

$$T = \frac{J \dot{\phi}^2}{2} + \frac{m}{2} (\dot{x}^2 + \dot{y}^2 + \dot{z}^2) = \frac{J \dot{\phi}^2}{2} + \frac{m}{2} (R^2 \dot{\varphi}^2 + p^2 (\dot{\varphi} - \dot{\phi})^2)$$

$$V = mgz = mg(z_0 + p(\varphi - \phi))$$

$$L = T - V = \frac{J \dot{\phi}^2}{2} + \frac{m}{2} (R^2 \dot{\varphi}^2 + p^2 (\dot{\varphi} - \dot{\phi})^2) - mg(z_0 + p(\varphi - \phi))$$

$$\psi: \frac{d}{dt}(m(R^2\dot{\psi} + r^2\dot{\psi} - r^2\dot{\phi})) - mgp = 0$$

$$R^2\ddot{\psi} + r^2\ddot{\psi} - r^2\ddot{\phi} - gp = 0 =$$

$$\phi: \frac{d}{dt}(J\dot{\phi} + m(r^2\dot{\phi} - r^2\dot{\psi})) + mgp = 0$$

$$J\ddot{\phi} + mp^2(\dot{\phi} - \dot{\psi}) + mgp = 0$$

$$\ddot{\phi} = \frac{mp^2\ddot{\psi} - mgp}{J + mp^2}$$

$$\ddot{\psi}(R^2 + p^2) - p^2 \frac{mp^2\ddot{\psi} - mgp}{J + mp^2} - gp = 0$$

$$\ddot{\psi} \left[R^2 + p^2 - \frac{mp^4}{J + mp^2} \right] = gp - \frac{mgp^3}{J + mp^2}$$

$$\ddot{\psi} = \frac{gp(J + mp^2) - mgp^3}{(R^2 + p^2)(J + mp^2) - mp^4} = \frac{Jgp}{R^2J + R^2mp^2 + Jp^2}$$